GPU & Cooling Systems – Assignment

Task 1: GPU Identification

- **Brand:** NVIDIA (GPU chip), on an ASUS graphics card
- **Model:** NVIDIA GeForce series, ASUS board
- **Type:** Dedicated GPU

It is a separate add-in card, not part of the CPU. It has a PCIe x16 connector, its own video memory chips, a power delivery section with an external power connector, and multiple display outputs (3 DisplayPort, 1 HDMI).



Figure 1: My GPU (cooler removed)

Task 2: Cooling System

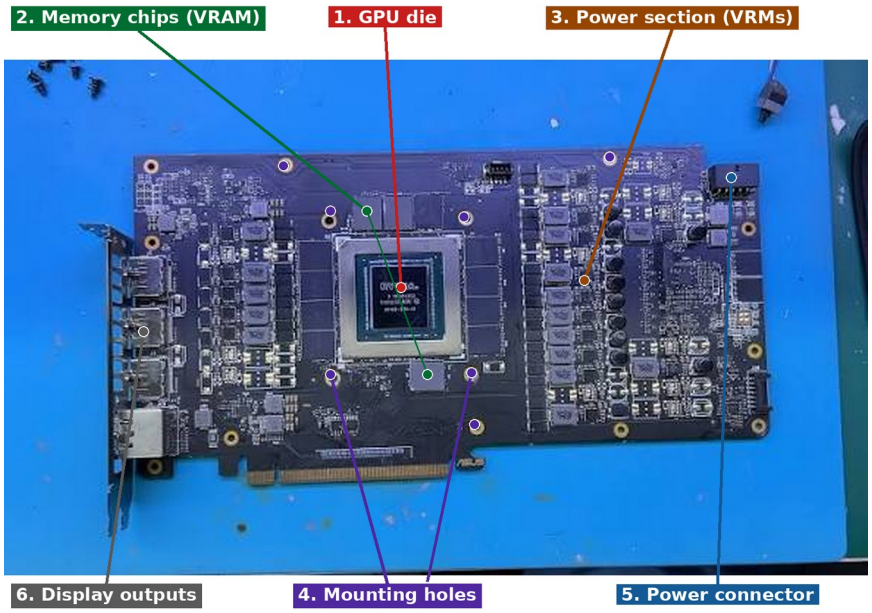


Figure 2: Labeled parts of my GPU

Labeled parts and their function:

- **GPU die:** The main heat source. The heatsink base presses on it.
- **Memory chips (VRAM):** Thermal pads carry their heat to the heatsink.
- **Power section (VRMs):** Regulates voltage and produces heat. It also uses thermal pads.
- **Mounting holes:** Screws here hold the heatsink tightly against the chip.
- **Power connector:** Brings extra power from the power supply to the card.
- **Display outputs:** Ports for monitors (3 DisplayPort, 1 HDMI).

Cooling parts that sit on top of this card when it is assembled:

- **Heatsink:** Metal fins and a copper base that absorb and spread heat.
- **Heat pipes:** Copper tubes that move heat from the die to the fins.
- **Fans:** Blow air through the fins to remove heat.
- **Thermal paste:** Fills gaps between the die and heatsink for better transfer.
- **Thermal pads:** Cool the memory and VRMs.

Task 3: Air vs Liquid Cooling

Factor	Air Cooling	Liquid Cooling
Cost	Cheaper, often included with the GPU	Costlier (pump, radiator, tubing)
Efficiency	Good for most GPUs, but louder and hotter under heavy load	Better heat removal, lower temps, quieter at full load
Maintenance	Almost none apart from dusting	Needs checks for leaks, pump wear and coolant
Installation	Simple, pre-installed	Complex, needs case space for a radiator
Reliability	Few failure points (fans only)	Pump failure or leaks could damage parts

Task 4: My Choice for an IPL Streaming and Gaming PC

I would choose air cooling. A good triple-fan air-cooled GPU handles graphics-heavy games and 1080p/1440p IPL streaming well, and it costs much less than liquid cooling. It needs almost no maintenance and has no leak risk, which suits long sessions during a match season. Liquid cooling mainly pays off for extreme overclocking or high-end cards, which most gamers don't need. The money saved can go toward a better GPU or good case airflow, which improves performance more than a costlier cooler.